Stolarsky monotonicity under logarithmic-mean smoothing

A solution of MathDB 376979

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Abstract

We prove both inequalities in Conjecture 16 of Horwitz's *Invariant Means* for every positive triple, with strictness unless all entries coincide. A fractional-linear normalization turns the question into two constrained extremum problems. Boundary estimates and signed-integral identities reduce both problems to triples with two equal entries. That remaining case follows from an explicit positive-coefficient certificate. The live MathDB record was checked on 11 September 2026 and reported open, with no posted claims.

1 Task 1: the problem and its solution

For $a, b > 0$ put $L(a, b) = (b - a)/(\log b - \log a)$, with $L(a, a) = a$, and set

$$T(a, b, c) = (L(b, c), L(c, a), L(a, b)).$$

Let $W = (W_1, W_2, W_3)$ be uniform on the two-simplex $\{w_i \geq 0 : \sum w_i = 1\}$, and define

$$F_q(a, b, c) = \mathbb{E}(W_1 a + W_2 b + W_3 c)^{-q}, \qquad M_q = F_q^{-1/q} \quad (q > 0).$$

The source's means are $U_0 = M_2$ and $U_1 = M_1$.

Theorem 1. *For every $a, b, c > 0$,*

$$M_2(T(a, b, c)) \geq M_2(a, b, c), \qquad M_1(T(a, b, c)) \leq M_1(a, b, c).$$

Equality in either inequality holds if and only if $a = b = c$.

The theorem answers both assertions in the source's Conjecture 16, including coincident coordinates. It also implies Conjecture 15's bounds for the invariant logarithmic mean. We count the present record as one problem; the latter consequence is not counted as a separate solve here.

2 Integral formulation

Writing $P(t) = (a + t)(b + t)(c + t)$, direct integration gives

$$F_2(a, b, c) = 2 \int_0^\infty \frac{dt}{P(t)}, \qquad F_1(a, b, c) = 2 \int_0^\infty \frac{t\, dt}{P(t)}.$$

These formulas remain nonsingular when coordinates coincide and are useful for both analysis and reliable numerical evaluation. For distinct coordinates they agree with the divided-difference formulas

$$F_2 = -2[\log](a, b, c), \qquad F_1 = 2[x \log x](a, b, c).$$

For completeness, integrating twice over the simplex gives $\mathbb{E}(X+t)^{-3} = 1/P(t)$, where $X = W_1a + W_2b + W_3c$. Now use $2\int_0^\infty (X+t)^{-3}\,dt = X^{-2}$ and $2\int_0^\infty t(X+t)^{-3}\,dt = X^{-1}$ and Tonelli's theorem.

3 A fractional-linear reduction of the general problem

Proposition 1. *Put $A = F_1(a,b,c)$, $B = F_2(a,b,c)$, $d = A/B$, and $P(t) = (a+t)(b+t)(c+t)$. Then*

$$T_i(a,b,c) = \frac{2}{A + Ba_i} = \frac{2/B}{a_i + d}.$$

The full pair of conjectured inequalities is equivalent to

$$P(d)\int_d^\infty \frac{dt}{P(t)} \geq d, \tag{1}$$

$$\left(\int_0^\infty \frac{dt}{P(t)}\right) P(d)\int_d^\infty \frac{t-d}{P(t)}\,dt \leq 1, \tag{2}$$

where $d = \int_0^\infty t/P(t)\,dt \big/ \int_0^\infty 1/P(t)\,dt$.

Proof. For $\{i,j,k\} = \{1,2,3\}$, the integral formulas yield

$$A + a_iB = 2\int_0^\infty \frac{dt}{(t+a_j)(t+a_k)} = \frac{2}{L(a_j,a_k)}.$$

The last identity also holds when $a_j = a_k$, by direct integration. For a positive triple r, substitution $t = 1/u$ shows

$$F_1(r^{-1}) = (r_1r_2r_3)F_2(r), \qquad F_2(r^{-1}) = (r_1r_2r_3)F_1(r).$$

Homogeneity therefore gives

$$F_1(T) = \frac{B}{2}P(d)F_2(a+d,b+d,c+d), \qquad F_2(T) = \frac{B^2}{4}P(d)F_1(a+d,b+d,c+d).$$

Insert the integral formulas and $A = dB$ to obtain (1) from $F_1(T) \geq F_1(a,b,c)$ and (2) from $F_2(T) \leq F_2(a,b,c)$. □

The next sections prove the general inequalities through a different normalization. The displayed tail inequalities are consequently true as well.

4 Complete proof when two coordinates coincide

Proposition 2. *If at least two of $a,b,c > 0$ coincide, both conjectured inequalities hold. Both are strict unless $a = b = c$.*

Proof. By symmetry and homogeneity it suffices to consider $(1,1,x)$, $x > 0$. Write $s = \log x$ and $\ell = (e^s - 1)/s$, with $\ell = 1$ at $s = 0$. Then $T(1,1,x)$ is a permutation of $(1,\ell,\ell)$. Direct integration gives

$$F_1(1,1,x) = \frac{2(x\log x - x + 1)}{(x-1)^2}, \qquad F_2(1,1,x) = \frac{2(x - 1 - \log x)}{(x-1)^2},$$

$$F_1(1,\ell,\ell) = \frac{2(\ell - 1 - \log \ell)}{(\ell-1)^2}, \qquad\qquad F_2(1,\ell,\ell) = \frac{2(\log \ell - 1 + 1/\ell)}{(\ell-1)^2}.$$

The two desired positive differences are identical:

$$F_1(1,\ell,\ell) - F_1(1,1,x) = F_2(1,1,x) - F_2(1,\ell,\ell) = \frac{2H(s)}{(\ell-1)^2},$$

where

$$H(s) = \frac{(\ell-1)^3}{s\ell^2} - \log \ell + 1 - \frac{1}{\ell}.$$

The apparent singularity at zero is removable and $H(0) = 0$. Differentiation gives

$$H'(s) = \frac{N(s, e^s)}{s^3(e^s-1)^3},$$

with the following polynomial in E:

$$\begin{aligned} N(s,E) = {} & (s-2)E^4 + (-s^3+s^2+8)E^3 \\ & + (s^4 - 2s^3 - 3s^2 - 6s - 12)E^2 \\ & + (s^4 + 3s^3 + 3s^2 + 8s + 8)E - s^2 - 3s - 2. \end{aligned}$$

We now certify that every nonzero Taylor coefficient of each entire function $N(s, e^s)$ and $-e^{4s}N(-s, e^{-s})$ is positive. It follows that $N(s, e^s) > 0$ for $s > 0$, and $N(s, e^s) < 0$ for $s < 0$. The denominator of H' is positive on both half-lines. Thus H decreases to zero on the negative half-line and increases from zero on the positive half-line. In particular $H(s) > 0$ for $s \neq 0$.

Here are complete algebraic certificates for the coefficient assertion. Let A_n and B_n be the nth derivatives at zero of the respective entire functions. They vanish for $0 \le n \le 8$, and direct integer calculation gives

n	A_n	B_n
9	2520	2520
10	46200	54600
11	489720	674520
12	3945480	6255480
13	26865696	48487296
14	163315152	332155824
15	915554640	2079277200

For $n \ge 5$, writing $(n)_j = n(n-1)\cdots(n-j+1)$, expansion of the four exponential terms gives

$$\begin{aligned} A_n = {} & 4^{n-1}(n-8) + 3^{n-3}\{-n(n-1)(n-5) + 216\} \\ & + 2^{n-4}P_2(n) + (n)_4 + 3(n)_3 + 3(n)_2 + 8n + 8, \\ P_2(n) = {} & n^4 - 10n^3 + 11n^2 - 50n - 192. \end{aligned}$$

For $n \ge 16$, $P_2(n) > 0$, since

$$P_2(m+16) = m^4 + 54m^3 + 1067m^2 + 9006m + 26400.$$

Also

$$4^{n-1}(n-8) > 3^{n-3}n(n-1)(n-5)$$

at $n = 16$, with difference 4380921872. The ratio of the left side to the right side increases thereafter: the required numerator is

$$4(n-7)(n-1)(n-5) - 3(n-8)(n+1)(n-4),$$

whose coefficients in $m = n - 16$ are $(1, 29, 288, 1044)$. This proves $A_n > 0$ for every $n \geq 16$.

For the second function the formula is

$$B_n = 4^{n-2}P(n) - 3^{n-4}Q(n) - 2^{n-4}R(n) - S(n),$$

where

$$\begin{aligned} P(n) &= n^2 - 13n + 32, \\ Q(n) &= n^4 - 15n^3 + 65n^2 - 267n + 648, \\ R(n) &= n^4 - 2n^3 - 13n^2 + 62n - 192, \\ S(n) &= n^3 - 2n^2 + n + 8. \end{aligned}$$

All four polynomials have positive coefficients after $n = m + 16$. At $n = 16$, the positive term exceeds the sum of the three subtracted terms by 12156288144. Each ratio of a subtracted term to the positive term strictly decreases for $n \geq 16$. To verify this last assertion, the coefficient lists in $m = n - 16$ of

$$4P(n+1)Q(n) - 3P(n)Q(n+1), \quad 4P(n+1)R(n) - 2P(n)R(n+1),$$

and $4P(n+1)S(n) - P(n)S(n+1)$, respectively, are

$$\begin{aligned} &(1, 64, 1677, 22884, 169766, 637568, 912000), \\ &(2, 162, 5410, 95486, 940476, 4905664, 10598400), \\ &(3, 200, 5273, 68572, 438232, 1094400). \end{aligned}$$

They are all positive. Hence $B_n > 0$ for every $n \geq 16$ as well. This completes the coefficient certificate and the proof. □

The companion script `verify_repeated_case.py` regenerates the derivative identity, the finite table, and all infinite-tail polynomial certificates using exact rational arithmetic. This restricted theorem is the input needed in the global extremum argument below; it is not counted separately.

5 Normalization to a constraint in the unit cube

For a positive triple v, abbreviate

$$A(v) = F_1(v), \qquad C(v) = F_2(v), \qquad P_v(t) = \prod_{i=1}^{3}(t + v_i).$$

For $p \in (0,1)^3$, set $q = \mathbf{1} - p$ and

$$K = \prod_i p_i, \quad g(p) = C(p) - C(q), \quad D(p) = A(p) + A(q) + C(p), \quad J(p) = KC(p)D(p).$$

Let $\mathcal{S} = \{p \in (0,1)^3 : g(p) = 0\}$.

Lemma 1. *For $v_i = (1-p_i)/p_i$ one has*

$$A(v) = KD(p), \qquad C(v) = K\{A(p) + A(q) + C(q)\}.$$

Consequently Theorem 1 is equivalent to

$$A(p) \geq 2, \qquad J(p) \leq 4 \quad (p \in \mathcal{S}).$$

Both inequalities are strict except at $p = (1/2, 1/2, 1/2)$.

Proof. For distinct coordinates, write $[f]_p$ for the second divided difference of f at the three coordinates of p. The change of nodes gives

$$[f]_v = K\,[xf((1-x)/x)]_p.$$

Since $A = 2[x \log x]$ and $C = -2[\log x]$, substituting these functions and expanding $\log((1-x)/x)$ gives the two asserted identities. They extend to all coordinates by continuity. In particular $A(v) - C(v) = Kg(p)$.

For an arbitrary original triple a, let $d = A(a)/C(a) > 0$ and $v = a/d$. Homogeneity gives $A(v) = C(v) = B$, say. Thus $p_i = 1/(1+v_i)$ belongs to $\mathcal{S}$ and $B = KD(p)$. The fractional-linear formula proved earlier now reads $T(v) = (2/B)p$. Hence

$$F_1(T(v)) = \frac{B}{2}A(p), \qquad F_2(T(v)) = \frac{B^2}{4}C(p).$$

The two desired inequalities are exactly $A(p) \geq 2$ and $J(p) \leq 4$. Conversely every $p \in \mathcal{S}$ comes from the normalized triple $v_i = (1-p_i)/p_i$. Equal coordinates correspond under this map, and the only constant triple on $\mathcal{S}$ is $(1/2, 1/2, 1/2)$. □

6 Boundary estimates

Lemma 2. *Along every sequence in $\mathcal{S}$ approaching the boundary of the unit cube,*

$$\liminf A(p) \geq 2, \qquad \limsup J(p) \leq 4.$$

Proof. Write $a = \min_i p_i$ and $c = \max_i p_i$. If $a \to 0$, then

$$C(p) \geq \frac{1}{2}\int_0^1 \frac{dt}{a+t} \longrightarrow \infty,$$

because the other two factors of $P_p(t)$ are at most 2 on $[0,1]$. Since $C(q) = C(p)$, this forces $\min_i q_i = 1 - c \to 0$; otherwise $C(q)$ would stay bounded. The converse follows by interchanging p and q. After passing to a convergent subsequence, every approach to the boundary therefore has $a \to 0$, $c \to 1$, and $C(p) = C(q) \to \infty$.

Coordinate monotonicity gives $A(p) \geq A(a, 1, 1) \to 2$, proving the first bound. Also $A(p)/C(p) \to 0$: for each fixed $\varepsilon > 0$, eventually $c \geq 1/2$ and

$$\frac{A(p)}{C(p)} \leq \varepsilon + \frac{2}{C(p)}\int_\varepsilon^\infty \frac{dt}{t(t+1/2)}.$$

The tail integral is finite. First take the limit and then let $\varepsilon \downarrow 0$. The same argument gives $A(q)/C(q) \to 0$. Thus $D(p)/C(p) \to 1$.

If $a \le b \le c$ are the coordinates of p, then

$$C(p) \le \frac{2}{c}\int_0^\infty \frac{dt}{(t+a)(t+b)} = \frac{2}{cL(a,b)} \le \frac{2}{c\sqrt{ab}}.$$

Here $L(a,b) \ge \sqrt{ab}$ follows immediately from $\sinh u/u \ge 1$. Consequently

$$KC(p)^2 \le \frac{4}{c}, \qquad J(p) = KC(p)^2\frac{D(p)}{C(p)} \quad \Longrightarrow \quad \limsup J(p) \le 4.$$

The subsequence argument proves the assertions for arbitrary boundary approaches, not just approaches to a specified face. □

7 Eliminating extrema with three distinct coordinates

Lemma 3. *No constrained local minimum of A on $\mathcal{S}$ has three distinct coordinates.*

Proof. Fix an interior point of $\mathcal{S}$, and write $P(u) = \prod_i (u+p_i)$ and $Q(u) = \prod_i (u+q_i)$. Differentiation under the integrals gives

$$\begin{aligned}
\partial_i A(p) &= -2\int_0^\infty \frac{t\,dt}{(t+p_i)P(t)},\\
\partial_i g(p) &= -2\int_0^\infty \left\{\frac{1}{(t+p_i)P(t)} + \frac{1}{(t+q_i)Q(t)}\right\} dt < 0.
\end{aligned}$$

Thus $\nabla g \ne 0$, and a constrained extremum satisfies $\partial_i A = \lambda \partial_i g$, with $\lambda > 0$ since both derivatives are negative. Equivalently,

$$\int_0^\infty \frac{(t-\lambda)\,dt}{(t+p_i)P(t)} - \lambda\int_0^\infty \frac{dt}{(t+q_i)Q(t)} = 0.$$

In the second integral put $u = -1-t$. Its denominator becomes $(u+p_i)P(u)$, since both factors change sign. With $R_i(u) = P(u)/(u+p_i)$, the three equations become

$$\int R_i(u)\,d\mu(u) = 0,$$

where the signed measure, supported on $(-\infty,-1)\cup(0,\infty)$, has density

$$\frac{d\mu}{du} = \begin{cases} -\lambda/P(u)^2, & u < -1,\\ (u-\lambda)/P(u)^2, & u > 0. \end{cases}$$

If the p_i are distinct, the three polynomials R_i span all polynomials of degree at most two: evaluation at $u = -p_j$ proves their linear independence. In particular $\int (u-\lambda)\,d\mu(u) = 0$. But its integrand is strictly positive on $u < -1$ and nonnegative on $u > 0$, with positive integral. All these integrals converge absolutely. This contradiction proves the lemma. □

Lemma 4. *No constrained local maximum of J on $\mathcal{S}$ has three distinct coordinates.*

Proof. Use $f = \log J$ and the same P, Q as above. The Lagrange equations give $\nabla f = \lambda \nabla g$. We first show $\lambda = 0$. The homogeneity and uniform-shift derivative identities are

$$\sum_i v_i \partial_i A(v) = -A(v), \quad \sum_i v_i \partial_i C(v) = -2C(v), \quad \sum_i \partial_i A(v) = -C(v), \quad \sum_i \partial_i C(v) = -\frac{2}{\prod_i v_i}.$$

The last two identities also follow directly by integration by parts. Applying them to $q = \mathbf{1} - p$ gives, on $g = 0$,

$$p \cdot \nabla D = -D, \qquad p \cdot \nabla f = 3 - 2 - 1 = 0, \qquad p \cdot \nabla g = -\frac{2}{Q(0)} < 0.$$

Taking the scalar product of the Lagrange equations with p therefore gives $\lambda = 0$, so all three partial derivatives of f vanish.

Writing $C = C(p)$ and $D = D(p)$, these equations are

$$\frac{1}{p_i} - 2\int_0^\infty \frac{1/C + (t+1)/D}{(t+p_i)P(t)}\,dt + \frac{2}{D}\int_0^\infty \frac{t}{(t+q_i)Q(t)}\,dt = 0.$$

The substitution $u = -1 - t$ in the last integral now expresses them as $\int R_i(u)\,d\nu(u) = 0$, where

$$d\nu(u) = \frac{1}{P(0)}\,d\delta_0(u) + \begin{cases} \dfrac{2(-1-u)}{DP(u)^2}\,du, & u < -1, \\ -\dfrac{2\{1/C + (u+1)/D\}}{P(u)^2}\,du, & u > 0, \end{cases}$$

and δ_0 denotes the unit point mass at zero. Indeed, $R_i(0)/P(0) = 1/p_i$. If the coordinates are distinct, the polynomials R_i span all polynomials of degree at most two. Thus $\int u\,d\nu(u) = 0$. In fact the integrand is strictly negative on both open intervals, while the point mass contributes zero. The integral is absolutely convergent and negative, a contradiction. □

8 Completion of the global proof

Proof of Theorem 1. Suppose $A(p) < 2$ somewhere on $\mathcal{S}$. A minimizing sequence for A has a subsequence converging in the closed cube. Lemma 2 excludes convergence to the boundary when the infimum is less than 2. The infimum is therefore attained at an interior point of $\mathcal{S}$. Lemma 3 forces two coordinates to coincide, contradicting Proposition 2 and the equivalence in Lemma 1. Hence $A(p) \geq 2$ on $\mathcal{S}$.

If $J(p) > 4$ somewhere on $\mathcal{S}$, Lemma 2 similarly forces its supremum to be attained in the interior. For clarity, it also ensures this supremum is finite: unbounded values on a compactness sequence could accumulate neither at the boundary nor at an interior point where J is continuous. Lemma 4 then forces two equal coordinates, again contradicting Proposition 2. Thus $J(p) \leq 4$ throughout $\mathcal{S}$.

An interior equality point for either established bound is itself a constrained extremum. The same lemmas force two equal coordinates there, and the strictness in Proposition 2 forces all three to equal $1/2$. Lemma 1 and homogeneity transfer these conclusions to every positive original triple, proving both inequalities and their equality cases. □

Consequence for the invariant mean. The source proves convergence of the logarithmic-mean iteration to (L_3, L_3, L_3). Along that iteration, the theorem makes U_0 increasing and U_1 decreasing. Continuity therefore gives $U_0(a,b,c) \leq L_3(a,b,c) \leq U_1(a,b,c)$. If the original triple is nonconstant, the first step is strict, so both bounds are strict as well.

9 Local behavior for three distinct coordinates

Proposition 3. *For fixed $q > 0$, put $a_i = 1 + u_i$, where $\sum_i u_i = 0$. As $u \to 0$,*

$$M_q(T(a)) - M_q(a) = \frac{3q-5}{96} \sum_i u_i^2 + \frac{19 - 4q - 3q^2}{480} \sum_i u_i^3 + O(\|u\|^4).$$

Consequently the two conjectured strict inequalities hold throughout some neighborhood of the diagonal, except on the diagonal itself. At $q = 5/3$, the difference takes both signs arbitrarily close to the diagonal.

Proof. Uniform simplex moments satisfy

$$\mathbb{E} W_1^i W_2^j W_3^k = \frac{2 i! j! k!}{(i+j+k+2)!}.$$

In particular, for $Z = \sum_i W_i u_i$ we have $\mathbb{E}Z = 0$, $\mathbb{E}Z^2 = \sum_i u_i^2/12$, and $\mathbb{E}Z^3 = \sum_i u_i^3/30$. Taylor expansion therefore gives

$$M_q(a) = 1 - \frac{q+1}{24} \sum_i u_i^2 + \frac{(q+1)(q+2)}{180} \sum_i u_i^3 + O(\|u\|^4).$$

For the logarithmic means, direct expansion of their reciprocal integral is

$$L(1+v, 1+w) = 1 + \frac{v+w}{2} - \frac{(v-w)^2}{12} + \frac{(v+w)(v-w)^2}{24} + O((|v| + |w|)^4).$$

Write μ for the arithmetic mean of $T(a)$ and $v_i = T_i(a) - \mu$. Using $\sum_i u_i = 0$, the preceding expansion yields

$$\begin{aligned} \mu &= 1 - \frac{1}{12} \sum_i u_i^2 + \frac{1}{24} \sum_i u_i^3 + O(\|u\|^4), \\ \sum_i v_i^2 &= \frac{1}{4} \sum_i u_i^2 - \frac{1}{4} \sum_i u_i^3 + O(\|u\|^4), \\ \sum_i v_i^3 &= -\frac{1}{8} \sum_i u_i^3 + O(\|u\|^4). \end{aligned}$$

Apply the expansion of M_q at mean μ, using homogeneity, and subtract the expansion at a. This proves the asserted coefficients. Analyticity on the positive octant makes the remainder uniform in direction near zero. The quadratic coefficients at $q = 1, 2$ are nonzero with the required signs. At $q = 5/3$, the cubic term is $\sum_i u_i^3/120$; taking $u = (t, t, -2t)$ and letting t approach zero from either side proves the final assertion. □

The exact symbolic coefficients are independently regenerated from the defining integrals by `verify_local_expansion.py`.

10 Follow-up tasks 2–5

Task 2. Classify the exponents $q > 0$ for which $M_q \circ T - M_q$ has a fixed global sign. The local expansion proves that a positive sign requires $q > 5/3$ and a negative sign requires $q < 5/3$; the critical exponent has both signs locally. Determine the full admissible parameter ranges.

Task 3. Determine the best constants in quantitative versions of the two inequalities under $\max(a,b,c)/\min(a,b,c) \le R$, normalizing each gap by $\sum_i (a_i - \bar{a})^2/\bar{a}$. Find their asymptotic behavior as $R \downarrow 1$ and $R \to \infty$.

Task 4. Extend the theorem to n variables. Define $M_q^{(n)}$ using the uniform $(n-1)$-simplex and put $(T_n(a))_i = M_1^{(n-1)}(a_1, \ldots, \widehat{a_i}, \ldots, a_n)$. Determine which exponents give global monotonicity under T_n, and whether the extremum reduction still reduces to repeated coordinates.

Task 5. Derive effective, certified error bounds for iteration of T toward the invariant logarithmic mean. Determine the sharp asymptotic contraction rate of the interval with endpoints $U_0(T^k(a,b,c))$ and $U_1(T^k(a,b,c))$, and give a stopping rule for any prescribed accuracy.

References

[1] A. Horwitz, *Invariant Means*, arXiv:math/0007095v1 (2000), Conjecture 16; journal version, J. Math. Anal. Appl. 270 (2002), 499–518. `https://arxiv.org/abs/math/0007095`.

[2] MathDB, problem 376979, live record retrieved 11 September 2026. `https://mathdb.com/p/376979`.